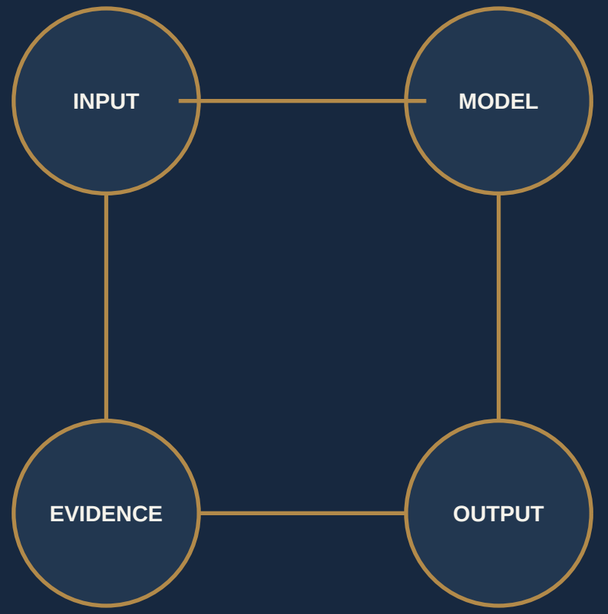
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Machine Learning

AI.103 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define supervised.
- Apply key methods including unsupervised and semi-supervised learning.
- Build or analyze a practical system involving regression.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Supervised

02

METHODS

Unsupervised and semi-supervised learning

03

SYSTEMS

Regression

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Supervised

Supervised

02 · Unsupervised and semi-superv

Unsupervised and semi-supervised learning

03 · Regression

Regression

04 · Classification

Classification

05 · Clustering

Clustering

06 · Dimensionality reduction

Dimensionality reduction

07 · Feature engineering

Feature engineering

08 · Model evaluation

Model evaluation

Core ideas, made explicit

AI.103 • Distinguish definitions, assumptions and evidence

- Supervised
- Unsupervised and semi-supervised learning
- Regression
- Classification
- Clustering

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

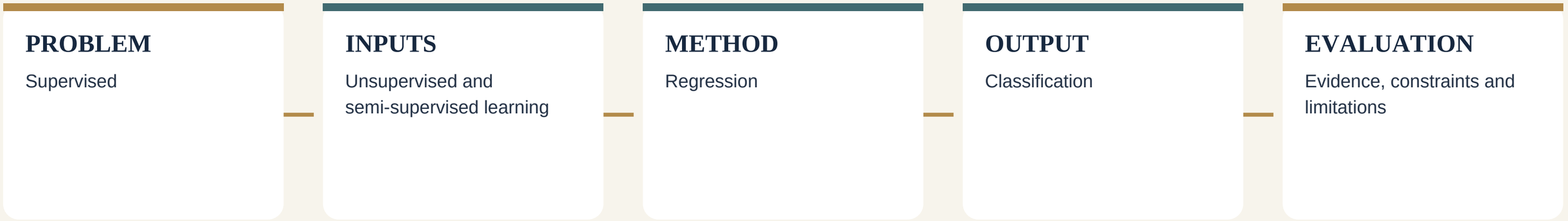
AI.103 • Distinguish definitions, assumptions and evidence

- Dimensionality reduction
- Feature engineering
- Model evaluation
- Overfitting
- Regularization

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.



Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving supervised.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using supervised.
- Implement or simulate a baseline using unsupervised and semi-supervised learning.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.